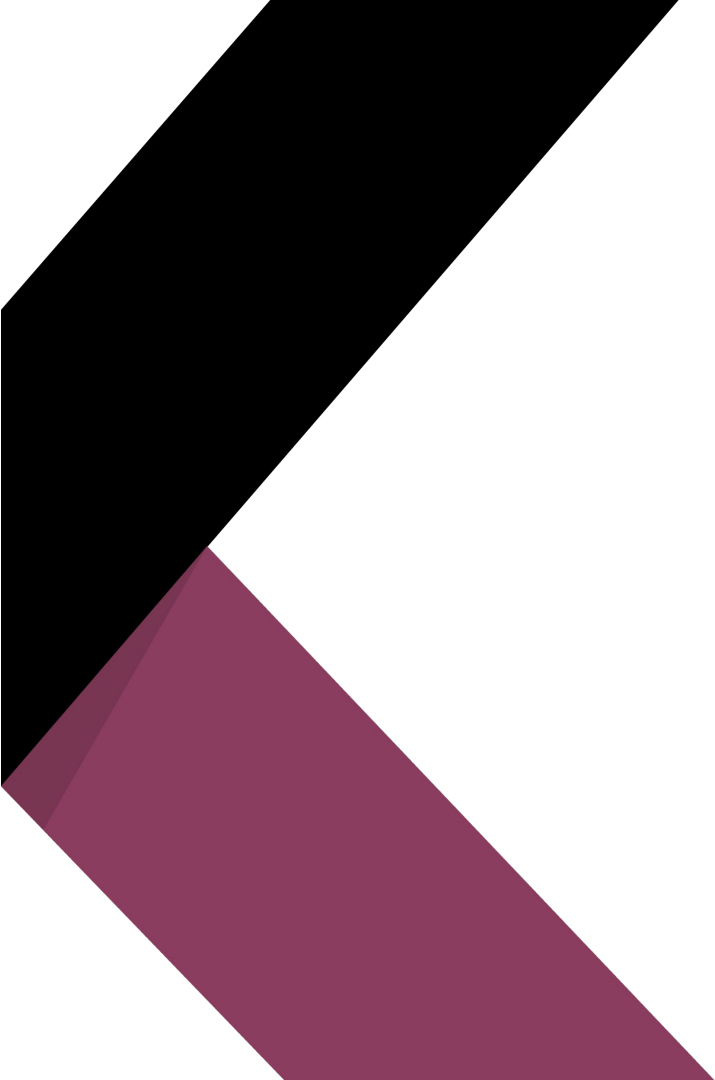




Induction machine



Introduction

Induction machine - Overview

- Commutatorless motor drives offer a number of advantages over conventional DC motor drives for electric propulsion, as low cost, high power factor, reliable operation.
- An AC induction motor is one of the simplest and most reliable types of electric AC motor, with the power ranging from a few watts to many kilowatts.
- There are many different ways to design and control such motors, from low cost with a single-phase input to high accuracy with a polyphase approach, and this type of system is the most widely used motor across the consumer and industrial markets.

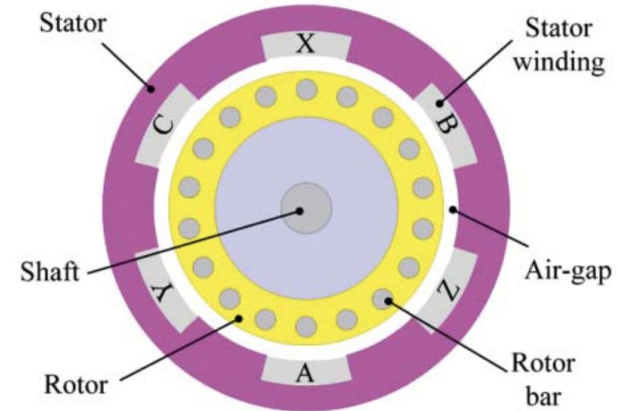
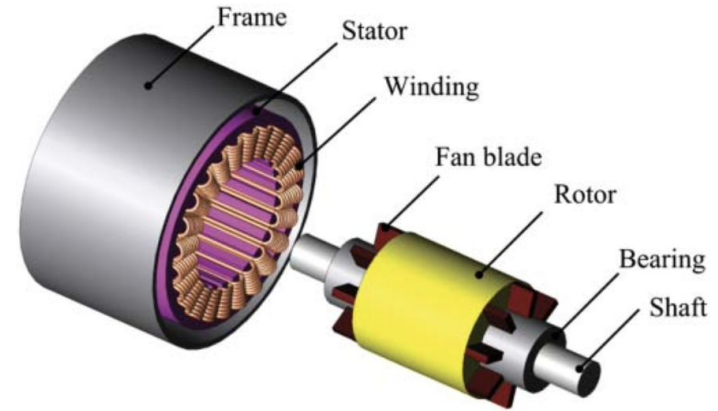


Induction motor – Structure

The most common type of IMs is the **squirrel-cage** IM.

It consists of:

- a **stator** incorporated with three-phase armature windings;
- a **rotor** incorporated with cage bars that are short-circuited by two end-rings;
- two end-**bearings** to support the rotor;
- a machine **frame** with two end-bells to house the machine.





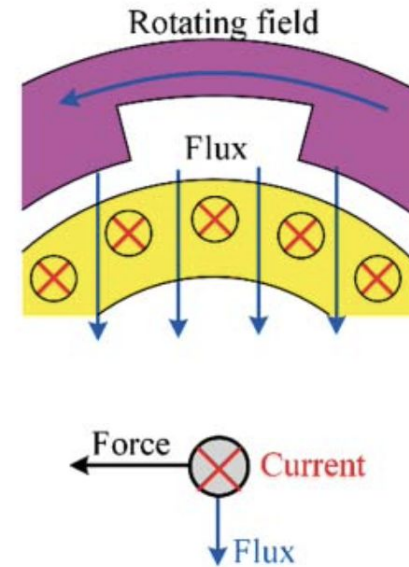
Induction motor – Working principle

The stator of a three-phase induction machine contains a series of windings to which three-phase electrical power is applied.

These windings establish **a rotating magnetic field** in the air gap between the stator and the rotor. The stator field can be seen as a series of north-south poles revolving around the circumference of the stator.

The rotation of the stator magnetic field occurs at a fixed speed called **synchronous speed** and linked to the supply frequency f .

In IMs, **the angular speed of the rotor is lower than the synchronous speed of the magnetic field** generated by the stator windings, hence called asynchronous.



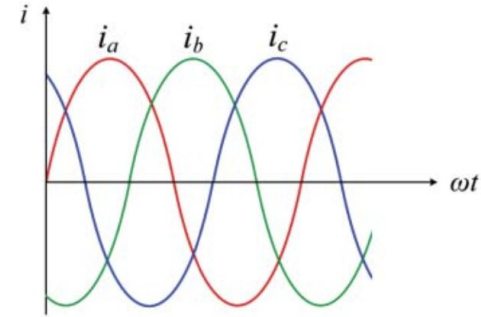
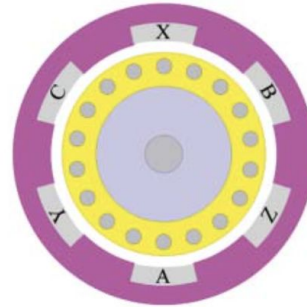


Induction motor – Working principle

The rotor windings are immersed in this rotating magnetic field. Since the rotor turns slower than the stator field, the magnetic flux linkage in the rotor windings varies.

The rotating magnetic field **induces currents in the rotor** by magnetic induction (Faraday's law).

These induced currents, in turn, **generate a rotor magnetic field**, which opposes flux variations; it interacts with the stator field, **generating a torque** on the rotor windings which causes the rotor to rotate.





Induction motor – Working principle

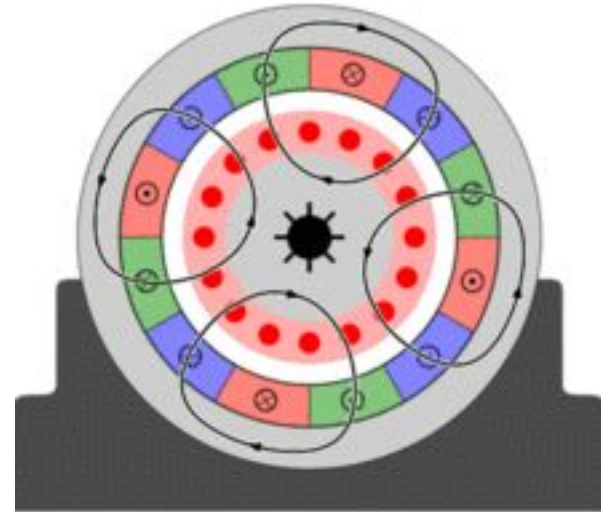
The link between synchronous speed, supply frequency f and the number p of poles (or p_c of pole pairs per phase) is expressed by the relation:

$$n_s = \frac{120f}{p} = \frac{60f}{p_c}$$

Since the relative motion between the rotating field and the rotor is the key for torque production, the **slip s** is defined as:

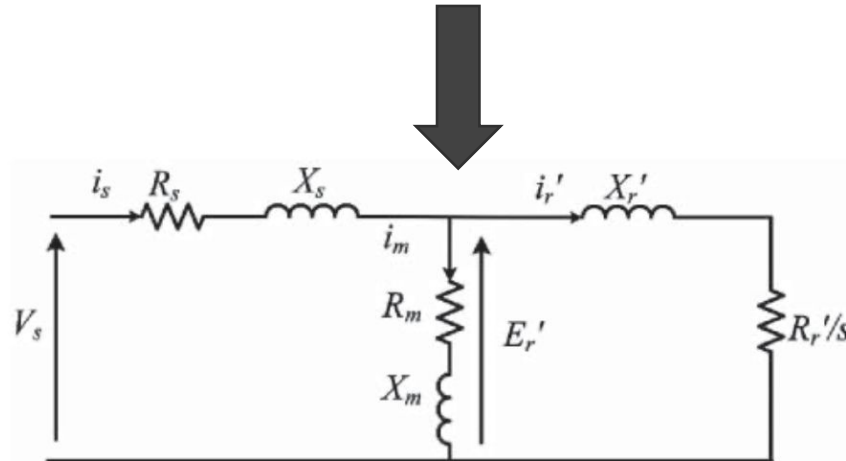
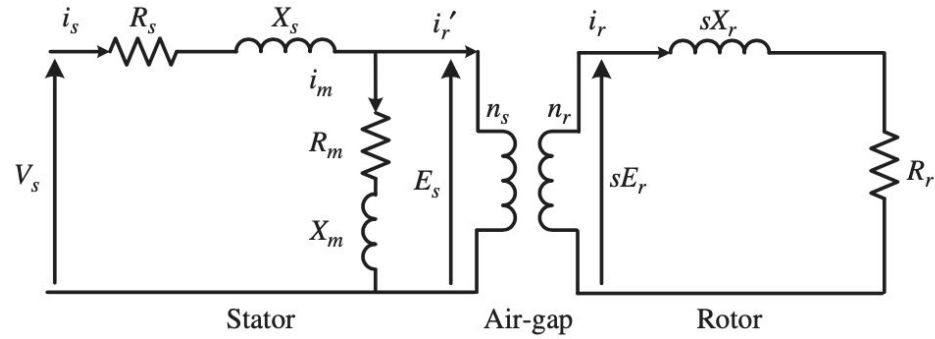
$$s = \frac{\omega_s - \omega_r}{\omega_s}$$

Where ω_s represents the synchronous speed and ω_r the rotor speed.



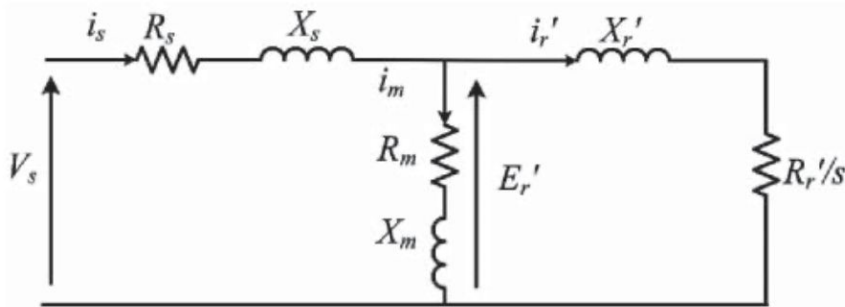


IM - Modeling





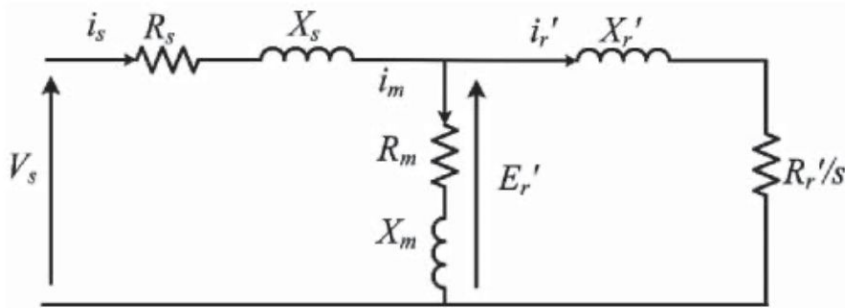
IM - Modeling



- R_s and X_s are respectively the winding resistance and leakage reactance in the stator;
- R_r and X_r are respectively the winding resistance and leakage reactance in the rotor at standstill;
- E_s and E_r are respectively the induced EMF in the stator and induced EMF in the rotor at standstill;
- R_m and X_m are respectively the core-loss resistance and magnetizing reactance in the stator;
- k is the ratio of the number of turns of stator winding to the number of turns of rotor winding.



Equivalent circuit - Equations



$$Z_s = R_s + jX_s \quad I_s = \frac{V_s}{(Z_s + Z_m || Z_r')}$$

$$Z_m = R_m + jX_m$$

$$Z_r' = \frac{R_r'}{s} + jX_r'$$

$$I_r' = \frac{Z_m}{Z_m + Z_r'} I_s$$

$$P_s = \frac{3R_r'}{s} I_r'^2$$

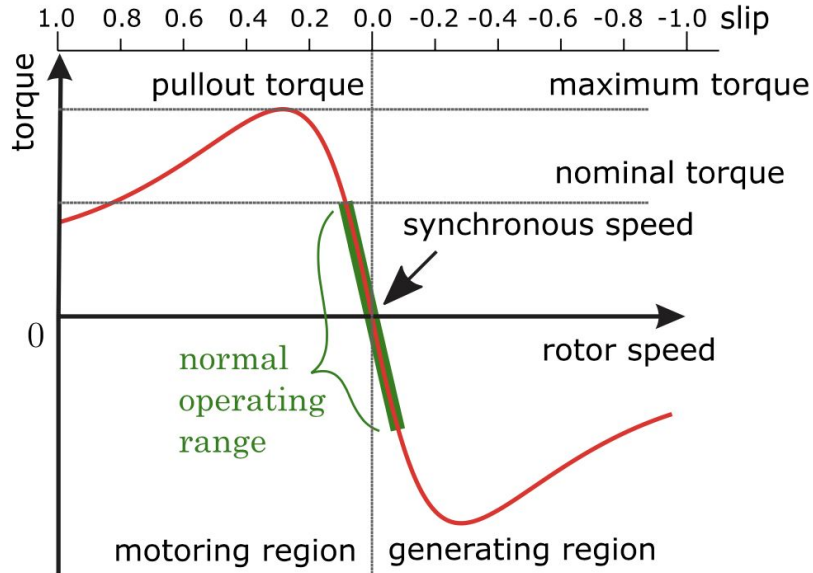
$$\Delta P_{Cu_r} = 3R_r' I_r'^2$$

$$P_m = P_s - \Delta P_{Cu_r} = 3 \frac{1-s}{s} R_r' I_r'^2$$

$$C = \frac{P}{\omega_r} P_m = \frac{3P}{\omega_r} R_2' \frac{1-s}{s} I_r' = \frac{3P}{\omega} \frac{R_2'}{s} I_r'^2$$



Torque – speed relationship

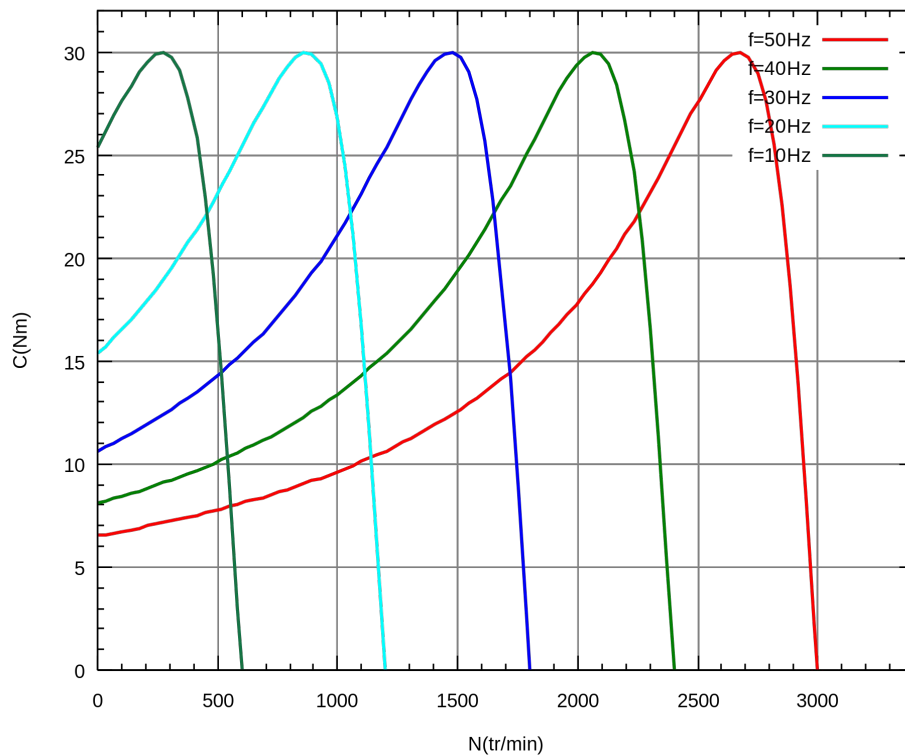


At synchronous speed, no torque is produced by the motor, because no current is induced in the rotor windings.

If the rotor rotates at a lower speed than the stator field, the motor runs in **motoring mode**. Otherwise, the motor operates in **generator mode**. The maximum torque, which can be obtained at the motor shaft, is called **pullout torque**. The nominal torque is conventionally half of the pullout torque. The motor usually operates in the **linear region** between the positive and the negative nominal torque.



Speed control

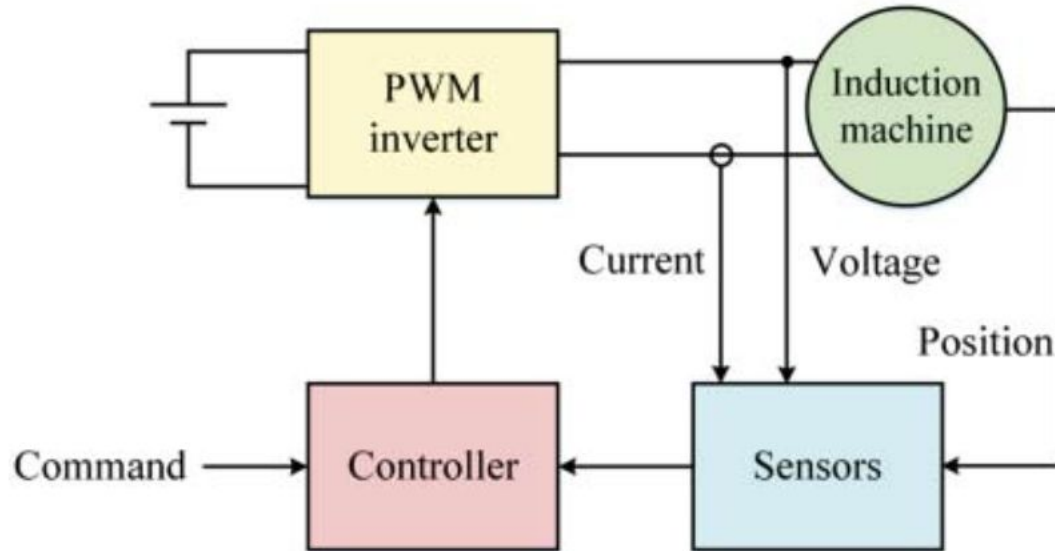




System configuration



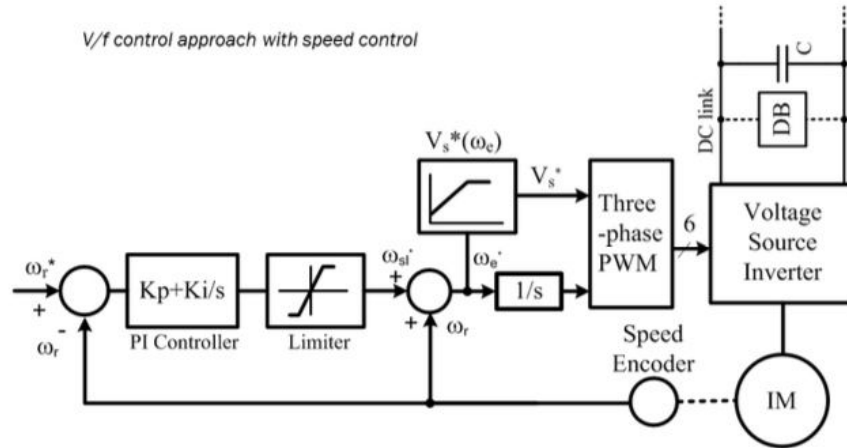
Induction motor drive





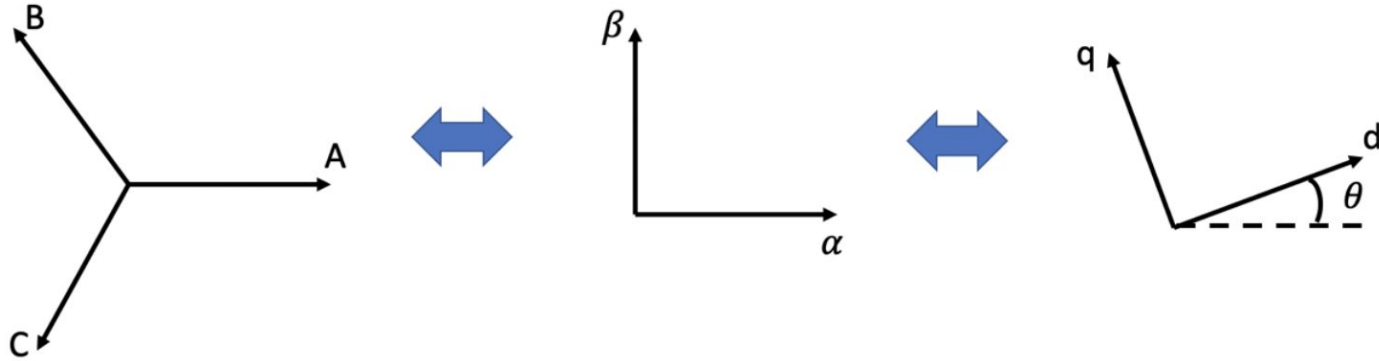
Control strategies – V/f

In the past, mainly the rather simple **voltage-to-frequency (V/f) control** was used and it can still be found in industrial drives. This method is a **scalar method**, that is, it operates just on the module (and not the phase) of the control quantities involved. The main drawback is its **limited dynamic performance**.





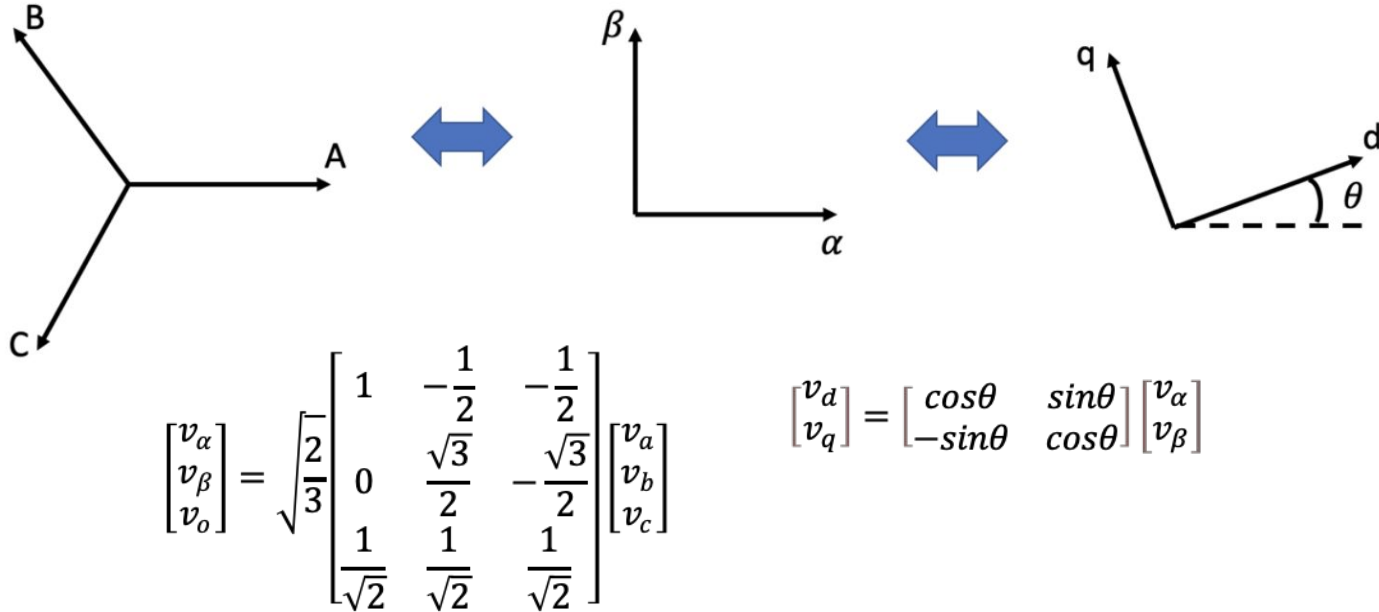
Park – Clarke transformations



With the letters **ABC** is denoted the three-phase stator power supply system, with $\alpha\beta$ the equivalent two-phase stator (and therefore stationary) reference system and with **dq** the reference system joined with the rotor and therefore rotating at rotor speed. The transformation from ABC to $\alpha\beta$ is called **Park transformation**, while $\alpha\beta$ to dq is called **Clarke transformation**.



Park – Clarke transformations





Control strategies - FOC

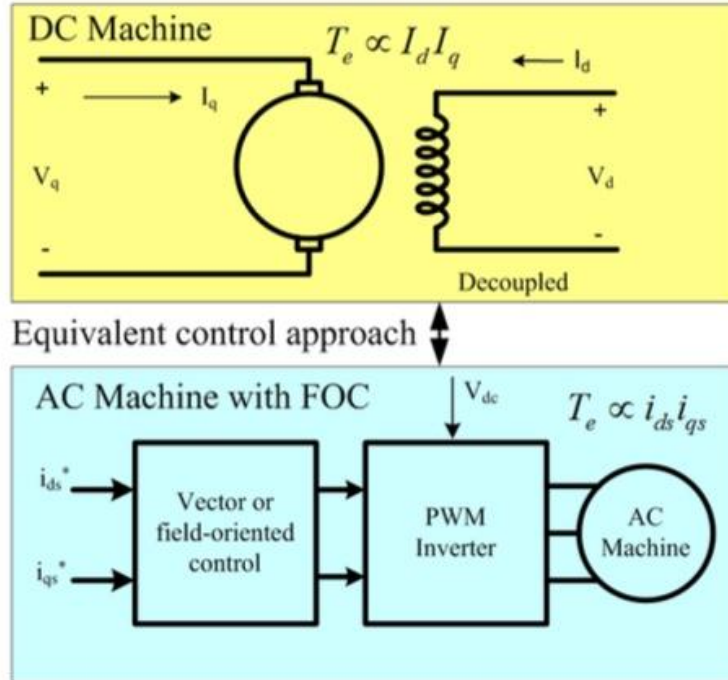
Another control strategy for inverter-fed variable-speed drives is the “**Field Oriented Control**” (FOC).

The time-dependent instantaneous three-phase system values, such as currents and voltages, are transformed into an orthogonal two-dimensional coordinate system by means of the Park transformation. This coordinate system rotates with the same frequency as the three-phase values.

From this, **the three AC quantities are reduced to two DC quantities** which, in general, are much simpler to control. The two DC quantities can be represented as a “space vector” in the two-dimensional coordinate system. This transformation results in a controllability of torque of the induction machine that is similar to the DC machine.



Control strategies - FOC



This technique controls the variable of interest **in module and phase** with the aim of separating, as occurs in the direct current machine, the current components that contribute to the flux and torque. With this type of regulation it is possible to obtain **rapid response and good adaptation** by the machine, whose operation, moreover, can be extended to the **four quadrants**.



Control strategies - FOC

The advantages of field-oriented control can be summarized as follows:

- **High efficiency** obtained by aligning the stator and rotor flows to generate excellent torque production by the machine;
- **High dynamic response** obtained by controlling the quantities (voltages, currents and fluxes) in module and phase;
- **Low torque ripple**, as the stator field is continuously updated according to the position of the rotor field;
- **Independent control of flux and torque**, as in direct current machines;
- **High dynamic range** through weakening operation;
- **High torque and low starting current.**



Inverter for induction machine

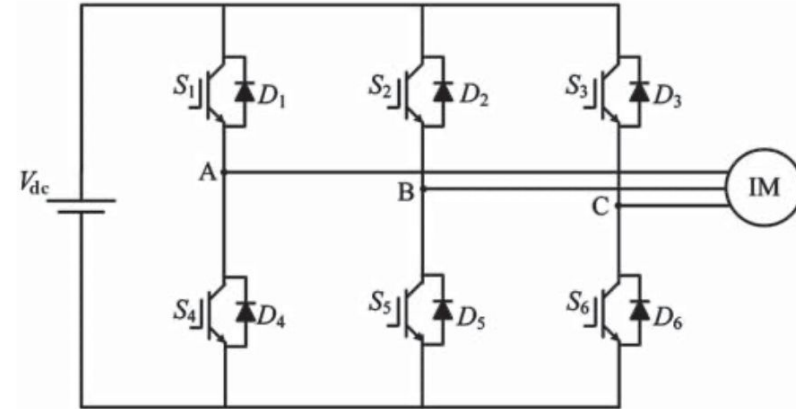


Inverter

Inverters for induction motors are generally classified into **voltage-fed** and **current-fed** types. Because of the need of a large series inductance to emulate a current source, the current-fed inverters are rarely used for EV propulsion. In fact, the voltage-fed inverters are almost exclusively used because they are very **simple** and can have **power flow in either direction**.

Control strategies can be classified as:

- **PWM** (Pulse Width Modulation);
- **Six-step**.

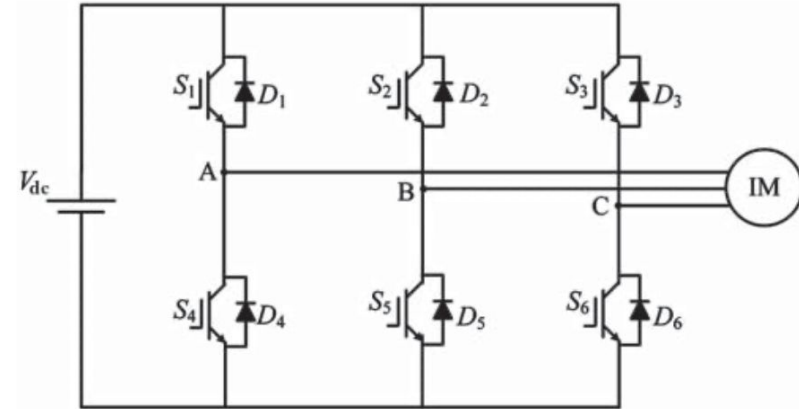




Inverter

The inverter design highly depends on the technology of power devices. Nowadays, the **insulated-gate bipolar transistor** (IGBT)-based inverter is most attractive and has been widely accepted by modern EVs. The selection of power devices is based on:

- **the voltage rating** is at least twice the nominal battery voltage;
- **the current rating** is large enough so that there is no need to connect several power devices in parallel;
- **the switching speed** is sufficiently high to suppress motor harmonics and acoustic noise.





Sinusoidal PWM control

The structure of the PWM inverter coincides with that of the 6-step inverter, what changes is the switching control technique of the various static switches.

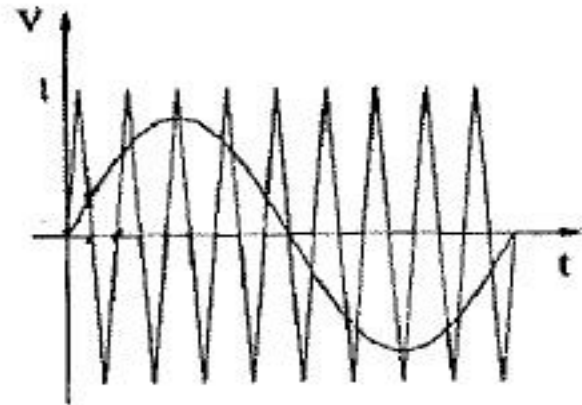
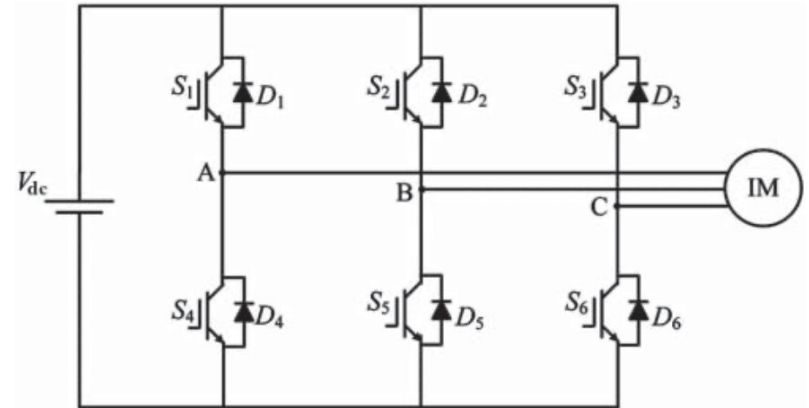
Comparison between:

- **Modulating sinusoid** of amplitude A_s and frequency f_s
- **Triangular carrier wave** of constant amplitude A_t and frequency f_t with:

$$A_t > A_s$$

$$f_t \gg f_s$$

Therefore, you do not get equal pulses, but **modulated in width (PWM)**.

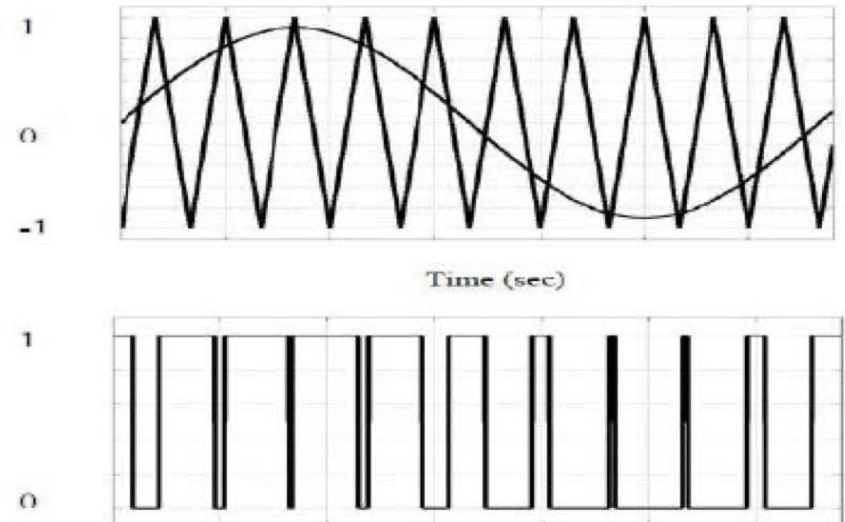




Sinusoidal PWM control

When the value assumed by one of the modulators exceeds that of the triangular carrier, the switch S of the corresponding phase is turned on while when the value assumed by one of the modulators is less than that of the triangular carrier it is turned off (and the other switch of the same branch is turned on).

The technique can be implemented either in an **analog** or **digital** way.

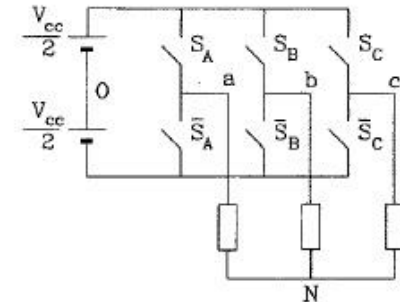
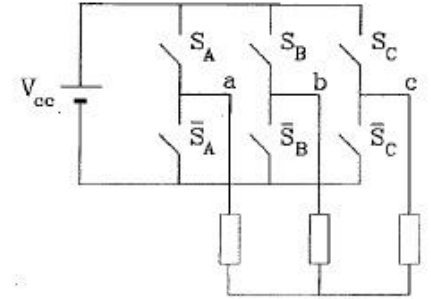
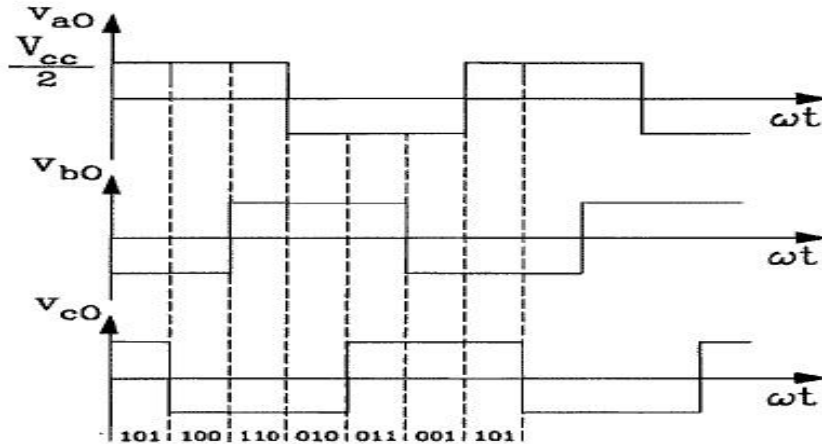




Six-step inverter control

In 6-step (or square wave) operation, the switch is alternately closed and open for a time equal to $T/2$, where T is the period of the alternating output voltage.

The switches of the same branch are controlled in a complementary way.





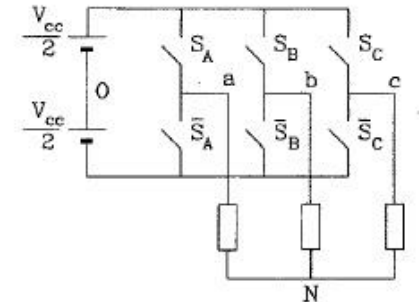
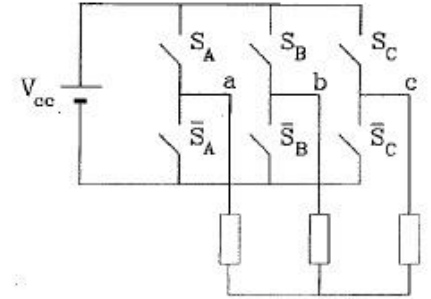
Six-step inverter control

In 6-step operation, each switch changes state every $T/2$, while the inverter changes state every $T/6$.

It is called a **6-step inverter** since the waveform is built in the 6 sixths of the period. Every sixth of a period, the components that are commanded for switching change.

Furthermore, in period T the inverter assumes 6 different states: (101), (100), (010), (011), (001). These are called **active states**.

The amplitude of the output voltage depends only on V_{CC} and the state of the switches.





Six-step inverter control

$$V_{ab} = V_{ao} - V_{bo}$$

$$V_{bc} = V_{bo} - V_{co}$$

$$V_{ca} = V_{co} - V_{ao}$$

$$V_{oN} = -\frac{1}{3}(V_{ao} + V_{bo} + V_{co})$$

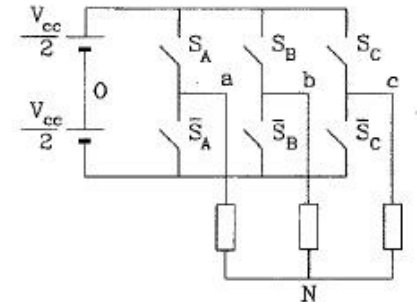
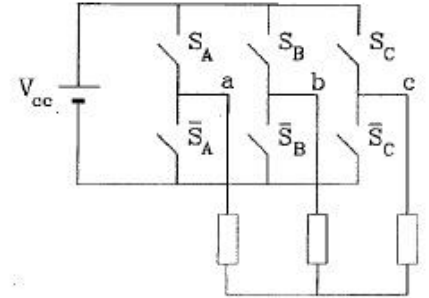
$$V_{aN} = V_{ao} + V_{oN}$$

$$V_{bN} = V_{bo} + V_{oN}$$

$$V_{cN} = V_{co} + V_{oN}$$

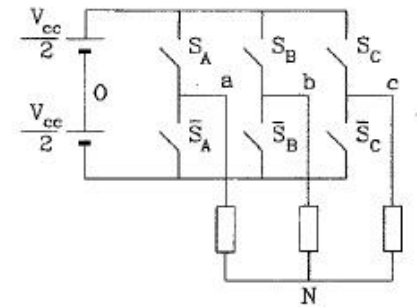
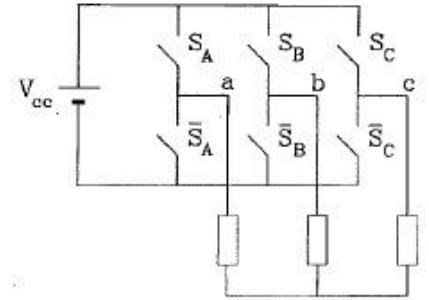
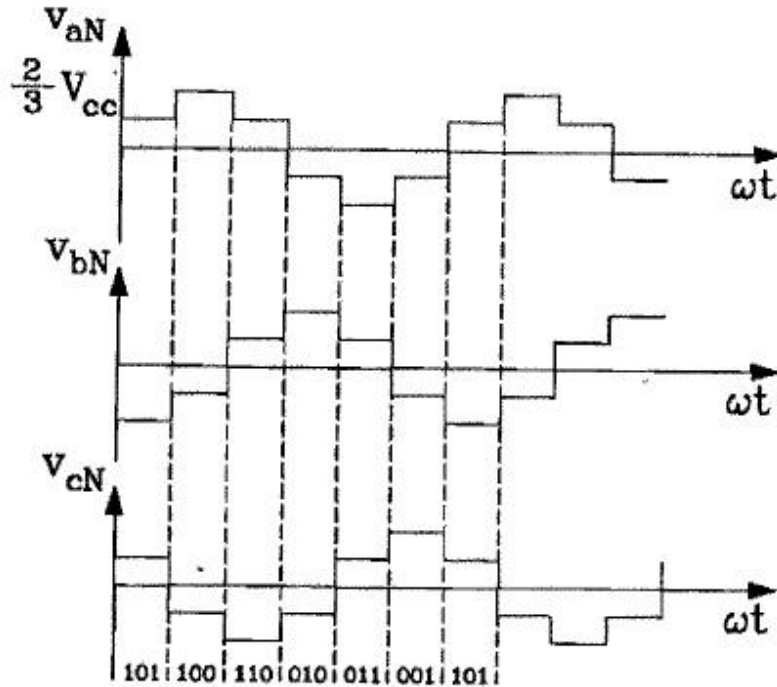
$$V_{aN} + V_{bN} + V_{cN} = 0$$

$$V_{ao} + V_{bo} + V_{co} + 3V_{oN} = 0$$





Six-step inverter control





Conclusions



Induction machine

Because of high cost, need for maintenance, and lack of sturdiness, the wound-rotor induction motor is less attractive than the squirrel-cage counterpart, especially for electric propulsion in electric vehicles (EVs). Hence, the **squirrel-cage** induction motor is loosely named as the induction motor for EV propulsion.

In addition to the common advantages of commutatorless motor drives, the induction motor drives possess the definite advantages of **low cost** and **ruggedness**. These advantages can generally outweigh their major disadvantage of **control complexity** and facilitate this technology being widely accepted for EVs.



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